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PEX TUBING EXPANDER HEAD

Abstract

An expansion toolhead configured to enable the expansion of multiple different sized diameters of cross-linked polyethylene (PEX) tubing with a single expansion toolhead. The expansion toolhead including a cap operably coupled to an expansion tool, and a plurality of jaws forming a 360 degree jaw section operably coupled to the cap, each of the plurality of jaws including an expansion tool interface portion forming a conical recess enabling a cone-shaped wedge of the expansion tool to move the jaw section between an unexpanded position and an expanded position, and a PEX tubing interface portion having a distal end, a proximal end and a plurality of lands positioned therebetween configured to form a corresponding plurality of substantially uniform diameter portions of the jaw section sequentially increasing in diameter between the distal end and the proximal end for the expansion of multiple different standard sized diameters of PEX tubing.

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Background/Summary

RELATED APPLICATION INFORMATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/629,020, filed Jan. 21, 2022, which is a National Phase entry of PCT Application No. PCT/US2020/043190, filed Jul. 23, 2020, which claims the benefit of U.S. application Ser. No. 16/519,684, filed Jul. 23, 2019, which issued as U.S. Pat. No. 11,110,646, the disclosures of which are fully incorporated herein by reference in their entirety.

TECHNICAL FIELD
[0002] The present disclosure relates generally to pipe and tubing expansion tools and methods, and more particularly to a cross-linked polyethylene (PEX) expansion toolhead configured to expand multiple different standard sized diameters of PEX tubing.
BACKGROUND
[0003] Cross-linked polyethylene, commonly known as PEX tubing, is a type of flexible plastic tubing often used in plumbing, geothermal and fire protection systems of residential and commercial construction. Frequently, PEX tubing, which generally costs about one-third of the equivalent copper pipe, is a substitute for copper and other more expensive plumbing options. Besides being less expensive, PEX tubing is also desirable because of its ease in installation.
[0004] One aspect that makes PEX tubing easy to install is its flexibility. As opposed to rigid pipe which generally comes in 10 foot sections, PEX tubing can be stored on a spool in lengths of up to 500 feet. During installation, PEX tubing can be uncoiled and pulled around corners and through offset openings, which would otherwise require a fitting or connection between sections of rigid pipe at each bend. Fewer fittings and connections translates to quicker installation (e.g., no soldering of joints at every corner), as well as fewer points of potential failure. As an added benefit, because of its flexibility, PEX tubing is resistant to fracturing if the liquid within the tubing should expand or freeze. For these reasons, today PEX tubing is installed in more new construction homes than copper and PVC tubing.
[0005] A fitting or connection between sections of PEX tubing is established by expanding one end of the tube, inserting a connector into the expanded portion of the tube, and retaining the connector in the expanded portion until a natural resiliency of the PEX tubing causes the expanded portion to shrink, thereby retaining the connector in position. Optionally, a clamping sleeve is expanded and shrunk over the tube end, in order to further enhance connection between the PEX tubing and connector. In some cases, the clamping sleeve can be expanded around the tube end at the same time as the tube end is expanded.
[0006] The ends of PEX tubing are typically expanded with an expander tool. Both manual and powered expander tools exist. One example of a manual PEX expander is described in U.S. Pat. No. 6,862,766, the contents of which are hereby incorporated by reference to the extent that they do not contradict the teachings herein. Common manufacturers of manual PEX expanders include Iwiss®, Ridgid® and Uponor®. One example of a powered PEX expander is described in U.S. Pat.

No. 8,517,715, the contents of which are hereby incorporated by reference to the extent that they do not contradict the teachings herein. Common manufacturers of powered PEX expanders include Milwaukee Tool® and DeWalt®.

[0007] PEX tubing comes in a variety of standard size diameters, ranging from ⅜ inch up to 3 inches. With manual and powered expanders, each size of PEX tubing has its own toolhead, specifically shaped and sized to expand the PEX tubing the amount necessary to create a fitting (e.g., an expansion in diameter of about 3/16 inch). These toolheads can be purchased individually, or as a set of toolheads. On average, each toolhead costs between about $45 and about $60. Typically the different sized toolheads are color-coded to aid in their identification (e.g., blue for ½ inch toolheads, green for ¾ inch toolheads, red for 1 inch toolheads, etc.).

[0008] Referring to FIGS. **1**A and **1**B, an expansion toolhead **50** is depicted in accordance with the prior art. Such expansion toolheads **50** are often threadably coupleable to an expansion tool **52**, such as a manual or powered expansion tool, via a cap **54**. Typically the expansion toolheads **50** include a plurality of jaws **56** forming a 360 degree jaw section **58**. Both manual and powered expansion tools **52** use a cone-shaped wedge **60** to expand the jaw section **58** between an unexpanded position (as depicted in FIG. **1**A) and an expanded position (as depicted in FIG. **1**B).

[0009] Manual PEX expanders generally include a ratcheting mechanism to incrementally expand the toolheads. Powered PEX expanders, by contrast, utilize a motor and cam system to rapidly expand and contract the toolhead during use. For example, some powered PEX expanders cycle the toolhead at about 60 expansions per minute. Additionally, to promote a more even expansion of the PEX tubing, powered PEX expanders often rotate the toolhead slightly on each retraction stroke, such that the toolhead rotates with respect to the PEX tubing while it is expanding and retracting.

[0010] In residential construction, a plumbing system will typically use a combination of ½ inch, ¾ inch and 1 inch PEX tubing. Accordingly, a plumber working on such a job would need to carry three different sized toolheads (i.e., a ½ inch, ¾ inch, and 1 inch toolhead). As a typical bathroom, kitchen or utility room frequently includes all three sizes of PEX tubing, the plumber must continually switch between toolheads, which requires unscrewing the toolhead currently on the expander, storing the toolhead in a clean place, locating the desired toolhead, and screwing the desired toolhead onto the expander.

[0011] Although the individual steps for removing and replacing a toolhead are not complicated, over the course of the day, the time spent switching toolheads can add up to a significant amount. As a further complication, because the toolheads are in contact with the inside of water lines, the toolheads must be kept clean and free from the dirt and grime common to the construction environment. Even worse, misplaced or lost toolheads in the often dark and cramped workspaces can force the temporary halt of work on the plumbing system. As a result, the productivity in the building of plumbing and other systems that use PEX tubing continues to suffer. Unfortunately, no advances to address these concerns have been made in nearly a decade.

SUMMARY OF THE DISCLOSURE

[0012] Embodiments of the present disclosure provide a single toolhead configured to enable the expansion of multiple different standard size diameters of PEX tubing, thereby eliminating the need to remove and replace the toolhead when expanding different sizes of PEX tubing. Accordingly, embodiments of the present disclosure enable the replacement of a set of toolheads with a single toolhead. In some embodiments, the toolhead can be threadably coupled as an aftermarket feature to existing expansion tools, such as expansion tools manufactured by Iwiss® and Ridgid®, Milwaukee Tool®, DeWalt® and Uponor®. In other embodiments, the toolhead can be included as an integral component of an expansion tool, without a threaded coupling portion, thereby enabling the construction of a more compact toolhead.

[0013] One embodiment of the present disclosure provides an expansion toolhead for an expansion tool. The expansion toolhead can include a cap operably coupled to the expansion tool, and a plurality of jaws forming a 360 degree jaw section operably coupled to the cap. Each of the plurality of jaws can include an expansion tool interface portion forming a conical recess enabling a cone-shaped wedge of the expansion tool to move the jaw section between an unexpanded position and an expanded position, and a cross-linked polyethylene (PEX) tubing interface portion having a distal end, a proximal end and a plurality of lands positioned therebetween configured to form a corresponding plurality of substantially uniform diameter portions of the jaw section sequentially increasing in diameter between the distal end and the proximal end for the expansion of multiple different standard size diameters of PEX tubing.

[0014] In one embodiment, the jaw section includes at least six jaws. In one embodiment, the PEX tubing interface portion of each of the plurality of jaws includes a roughened surface configured to grip an inside of the PEX tubing during expansion. In one embodiment each of the plurality of jaws includes a magnet configured to inhibit separation of the jaw section from the cone-shaped wedge. In one embodiment, each of the plurality of jaws includes one or more biasing members at least partially encircling the jaw section to bias the jaw section to the unexpanded position. In one embodiment each of the plurality of jaws includes a pair of biasing members. In one embodiment each of the plurality of jaws includes an expansion tool slot configured to interface with a portion of the expansion tool to at least partially rotate the jaw section relative to the expansion tool during operation.

[0015] In one embodiment, each of the plurality of jaws includes at least two lands sized for the expansion of at least two different standard size diameters of PEX tubing. In one embodiment, a first land is sized for the expansion of ½ inch PEX tubing, and a second land is sized for the expansion of ¾ inch PEX tubing. In one embodiment, each of the plurality of jaws includes a first ramp portion positioned in proximity to the distal end. In one embodiment, each of the plurality of jaws includes a second ramp portion positioned between the first land and the second land. In one embodiment, the first ramp portion has a steeper angle with respect to a centerline of the toolhead than the second ramp portion. In one embodiment, the first ramp portion and the second ramp portion have substantially the same angle with respect to a centerline of the toolhead.

[0016] In one embodiment, each of the plurality of jaws includes at least three lands sized for the expansion of at least three different standard size diameters of PEX tubing. In one embodiment, a first land is sized for the expansion of ½ inch PEX tubing, a second land is sized for the expansion of ¾ inch tubing, and a third land is sized for the expansion of 1 inch PEX tubing. In one embodiment, each of the plurality of jaws includes a first ramp portion positioned in proximity to the distal end. In one embodiment, each of the plurality of jaws includes a second ramp portion positioned between the first land and the second land, and a third ramp portion positioned between the second land and the third land. In one embodiment, the first ramp portion has a steeper angle with respect to a centerline of the toolhead than the second ramp portion, and the second ramp portion has a steeper angle with respect to a centerline of the toolhead than the third ramp portion. In one embodiment, the first ramp portion, the second ramp portion, and the third ramp portion have substantially the same angle with respect to a centerline of the toolhead.

[0017] The summary above is not intended to describe each illustrated embodiment or every implementation of the present disclosure. The figures and the detailed description that follow more particularly exemplify these embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The disclosure can be more completely understood in consideration of the following detailed description of various embodiments of the disclosure, in connection with the accompanying drawings, in which:

[0019] FIG. **1**A is a partial, cross-sectional view depicting an expansion toolhead operably coupled

to an expansion tool in an unexpanded position, in accordance with the prior art.
[0020] FIG. **1**B is a partial, cross-sectional view depicting the expansion toolhead of FIG. **1**A in an expanded position.
[0021] FIG. **2**A is a partial, cross-sectional view depicting an expansion toolhead operably coupled to an expansion tool in an unexpanded position, in accordance with an embodiment of the disclosure.
[0022] FIG. **2**B is a partial, cross-sectional view depicting the expansion toolhead of FIG. **2**A in an expanded position.
[0023] FIG. **3** is a perspective view depicting a proximal end of an expansion toolhead, in accordance with an embodiment of the disclosure.
[0024] FIG. **4**A is a perspective view depicting an expansion toolhead having two lands, in accordance with an embodiment of the disclosure.
[0025] FIG. **4**B is a profile view depicting the expansion toolhead of FIG. **4**A.
[0026] FIG. **4**C is an end view depicting the expansion toolhead of FIG. **4**B.
[0027] FIG. **5**A is a perspective view depicting an expansion toolhead having three lands, in accordance with an embodiment of the disclosure.
[0028] FIG. **5**B is a profile view depicting the expansion toolhead of FIG. **5**A.
[0029] FIG. **5**C is an end view depicting the expansion toolhead of FIG. **5**B.
[0030] FIG. **6**A is an expansion toolhead including ramps of an equal angle with respect to a centerline of the toolhead, in accordance with an embodiment of the disclosure.
[0031] FIG. **6**B is an expansion toolhead including ramps of an increasing angle with respect to a centerline of the toolhead, in accordance with an embodiment of the disclosure.
[0032] FIG. **6**C is an expansion toolhead including a combination of risers and ramps, in accordance with an embodiment of the disclosure.
[0033] FIG. **6**D is a perspective view depicting the expansion toolhead of FIG. **6**C, in an unexpanded position, in accordance with an embodiment of the disclosure.
[0034] FIG. **6**E is a perspective view depicting the expansion toolhead of FIG. **6**C, in an expanded position, in accordance with an embodiment of the disclosure.
[0035] FIG. **7**A is an expansion toolhead including ramps of an equal angle with respect to a centerline of the toolhead, in accordance with an embodiment of the disclosure.
[0036] FIG. **7**B is an expansion toolhead including ramps of an increasing angle with respect to a centerline of the toolhead, in accordance with an embodiment of the disclosure.
[0037] FIG. **7**C is an expansion toolhead including a combination of risers and ramps, in accordance with an embodiment of the disclosure.
[0038] FIG. **7**D is a perspective view depicting the expansion toolhead of FIG. **7**C, in an unexpanded position, in accordance with an embodiment of the disclosure.
[0039] FIG. **7**E is a perspective view depicting the expansion toolhead of FIG. **7**C, in an expanded position, in accordance with an embodiment of the disclosure.
[0040] FIG. **8**A is an expansion toolhead operably coupled to a manual expansion tool, in accordance with an embodiment of the disclosure.
[0041] FIGS. **8**B-D are expansion toolheads operably coupled to powered expansion tools, in accordance with embodiments of the disclosure.
[0042] FIG. **9**A is an expansion toolhead fixedly coupled to a manual expansion tool, in accordance with an embodiment of the disclosure.
[0043] FIG. **9**B is an expansion toolhead fixedly coupled to a powered expansion tool, in accordance with an embodiment of the disclosure.
[0044] While embodiments of the disclosure are amenable to various modifications and alternative forms, specifics thereof shown by way of example in the drawings will be described in detail. It should be understood, however, that the intention is not to limit the disclosure to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents,

and alternatives falling within the spirit and scope of the subject matter as defined by the claims.

DETAILED DESCRIPTION

[0045] Referring to FIGS. **2**A-B, a cross-sectional view of an expansion toolhead **100** is depicted in accordance with an embodiment of the disclosure. The expansion toolhead **100** can include a cap **102** and a plurality of jaws **104**. In one embodiment, the cap **102** can include a proximal end **106**, a distal end **108** and a wall **110** traversing therebetween defining an inner aperture **112**. In some embodiments, the inner aperture **112** can include a threaded portion **114** in proximity to the proximal end **106**, thereby enabling the expansion toolhead **100** to be selectively coupleable to an expansion tool **116**. For example, in one embodiment, the expansion toolhead **100** can be threadedly coupled to one of the manual or powered expansion tools described in the background section. In other embodiments, the cap **102** can be fixedly coupled to the expansion tool **116**, such that the expansion toolhead **100** is included as an integral component of the expansion tool **116**.

[0046] The plurality of jaws **104** can collectively form a 360 degree jaw section **118**. In one embodiment, the jaw section **118** includes at least six jaws **104**, although other numbers of jaws are also contemplated. For example, in one embodiment, the expansion toolhead **100** can include as few as two jaws **104** or as many as ten or more jaws **104**. Each of the plurality of jaws **104** can include a proximal end **120** and a distal end **122**. The plurality of jaws **104** can be operably coupled to the cap **102** in proximity to their respective proximal end **120**. For example, in one embodiment, the inner aperture **112** of the cap **102** can define a radial slot **124** configured to receive a radial extension **126** of each of the plurality of jaws **104**.

[0047] Each of the plurality of jaws **104** can include an expansion tool interface portion **128** forming a conical recess **130**, thereby enabling a cone-shaped wedge **132** of the expansion tool **116** to move the jaw section **118** between an unexpanded position (as depicted in FIG. **2**A) and an expanded position (as depicted in FIG. **2**B). In one embodiment, each of the plurality of jaws **104** can include a magnet **134** configured to inhibit separation of the jaw section **118** from the cone-shaped wedge **132**. In one embodiment, a biasing member **136**, such as an elastic band or spring, can at least partially encircle the jaw section **118** at a proximal end **120** of the plurality of jaws **104**, thereby biasing the jaw section **118** to the unexpanded position. In one embodiment, a biasing member **138** can be positioned distally along the jaw section **118** and at least partially encircle the jaw section **118**, thereby biasing the jaw section **118** to the unexpanded position. Various combinations of the magnet **134** and biasing members **136**, **138** are contemplated.

[0048] Referring to FIG. **3**, the proximal end **120** of the expansion toolhead **100** is depicted in accordance with an embodiment of the disclosure. In one embodiment, each of the plurality of jaws **104** includes an expansion tool slot **140** configured to interface with a portion of the expansion tool **116** to at least partially rotate the jaw section **118** relative to the expansion tool **116** during operation. For example, in one embodiment, the expansion tool slot **140** can be defined on the proximal end **120** of the radial extension **126** of each of the plurality of jaws **104**. In some embodiments, manipulation of the expansion tool slot **140** can cause the expansion toolhead **100** to incrementally rotate in either a clockwise or counterclockwise rotation during each expansion and contraction cycle of the expansion toolhead **100**. In other embodiments, manipulation of the expansion tool slot **140** can cause the expansion toolhead **100** to ratchet back and forth between a clockwise and counterclockwise rotation during each expansion and contraction cycle of the expansion toolhead **100**.

[0049] Referring to FIGS. **4**A-C, in one embodiment, each of the plurality of jaws **104** can include a PEX tubing interface portion including a plurality of lands **142**A, **142**B configured to form a corresponding plurality of substantially uniform diameter portions **144**A, **144**B of the jaw section **118**. In one embodiment, the substantially uniform diameter portions **144**A, **144**B can vary in their uniformity slightly across their length, thereby allowing for some tolerance in diameter variability. The plurality of substantially uniform diameter portions **144**A, **144**B can sequentially increase in diameter between the distal end **122** and the proximal end **120** for the expansion of multiple different standard sized diameters of PEX tubing. Accordingly, embodiments of the present disclosure provide a single toolhead **100** configured to enable the expansion of multiple different standard sized diameters of PEX tubing, thereby eliminating the need to remove and replace toolheads when expanding different sizes of PEX tubing.

[0050] As depicted in FIGS. **4**A-C, in one embodiment, each of the plurality of jaws **104** can include at least two lands **142**A, **142**B for the expansion of at least two different standard sized diameters of PEX tubing. In one embodiment, the first land **142**A can be sized for the expansion of ½ inch PEX tubing, and the second land **142**B can be sized for the expansion of ¾ inch tubing. Alternatively, the first land **142**A can be sized to expand PEX tubing having a size designation of between ⅜ inch and 1 inch, and the second land **142**B can be sized to expand PEX tubing having any larger size designation.

[0051] It is recognized that the measurable internal and external diameters of PEX tubing may differ slightly from its given size designation (e.g., the internal/external diameter of ⅜ inch PEX tubing may not measure exactly ⅜ inch). Embodiments of the present disclosure preferably are compatible with standard sized PEX tubing designations (e.g., ⅜, inch, ½ inch, ⅝ inch, ¾ inch, 1 inch, 1¼ inch, etc.) regardless of the actual dimensions of the PEX tubing. It is also contemplated that the expansion toolhead **100** can be utilized to expand other types of resilience tubing, hose and pipe, which may or may not conform to the designated sizes of PEX tubing.

[0052] Referring to FIGS. **5**A-C, in another embodiment of an expansion toolhead **200**, each of the plurality of jaws **104** can include at least three lands **142**A, **142**B, and **142**C configured to form a corresponding plurality of uniform diameter portions **144**A, **144**B, and **144**C of the jaw section **118** for the expansion of at least three different standard sized diameters of PEX tubing. In one embodiment, the first land **142**A can be sized for the expansion of ½ inch PEX tubing, the second land **142**B can be sized for the expansion of ¾ inch tubing, and the third land **142**C can be sized for the expansion of 1 inch tubing. Alternatively, the first land **142**A can be sized to expand PEX tubing having a single size designation, and the second and third lands **142**B, **142**C can be sized to expand PEX tubing having any sequential or non-sequential larger size designations.

[0053] In some embodiments, a gradual transition can be positioned between the distal end **122** of the jaw section **118** and the first land **142**A, as well as between the various lands **142**A, **142**B and **142**C, thereby enabling a more gradual expansion of the PEX tubing during expansion operations. For example, referring to FIG. **6**A, in one embodiment, each of the plurality of jaws **104** can include a first ramp portion **146**A positioned in proximity to the distal end **122** of the expansion toolhead **300**. In particular, the first ramp portion **146**A can provide a gradual transition between a distal end **122** of the expansion tool **300** and the first land **142**A, thereby enabling an insertion of the toolhead **300** into the end of PEX tubing when an inner diameter of the PEX tubing is smaller than the diameter of the unexpanded first land **142**A.

[0054] In one embodiment, each of the plurality of jaws **104** can include a second ramp **146**B positioned between the first land **142**A and the second land **142**B. Where a third land **142**C is present, each of the plurality of jaws **104** can include a third ramp **146**C positioned between the second land **142**B and the third land **142**C. Additional ramps can be included where one or more subsequent lands are present, thereby providing a gradual transition up to the subsequent land.

[0055] In some embodiments, portions of the various ramps **146**A, **146**B and **146**C can have a roughened surface configured to grip inside of the PEX tubing during expansion. For example, in one embodiment, one or more of the ramps **146**A, **146**B and **146**C can include a plurality of ribs **147** circumferentially inscribed into a portion of the exterior of the jaw section **118**, thereby inhibiting PEX tubing from being inadvertently pushed off of the expansion toolhead during expansion operations. Other roughened surfaces, such as stippling, surface treatments or coatings are also contemplated. In one embodiment, the roughened surface can further be included on the various lands **142**A, **142**B and **142**C.

[0056] As depicted in FIG. **6**A, in some embodiments, the first ramp **146**A, second ramp **146**B and third ramp **146**C can all share the same pitch or angle with respect to a centerline of the toolhead. In other embodiments, the pitch or angle with respect to a centerline of the toolhead can vary between the various ramps **146**A, **146**B and **146**C. For example, in order to minimize the larger expansion forces generally required to expand larger diameter/sizes of PEX tubing, the ramps can be made more gradual (e.g., at a shallower angle) in proximity to larger diameter lands. A similar two-step embodiment is depicted in FIG. **7**A.

[0057] As depicted in FIG. **6**B, the first ramp **146**A can have a steeper pitch than the second ramp **146**B, and a second ramp **146**B can have a steeper pitch than the third ramp **146**C. One notable advantage to this approach is that it tends to reduce the overall length of the expansion toolhead **400**, thereby reducing the bulk and weight of the device, which presents both lower-cost manufacturing and case-of-use advantages. In other embodiments, the ramps can have a variable pitch or angle with respect to a centerline of the toolhead, which can be based in part on the expansion forces generally required to expand PEX tubing of a given diameter. A similar two-step embodiment is depicted in FIG. **7**B.

[0058] In order to further reduce the overall length of the expansion tool, in yet another embodiment, a combination riser and ramp can be provided between the various lands. For example as depicted in FIGS. **6**C-E, a riser **148**B can be positioned immediately proximal to the first land **142**A. Second ramp **146**B can be positioned proximally to the riser **148**B to provide a gradual transition to the second land **142**B. Similarly, a riser **148**C can be positioned immediately proximal to the second land **142**B, and the third ramp **146**C can be positioned proximally to the riser **148**C to provide a gradual transition to the third land **142**C. In some embodiments, the distal end **122** of the toolhead **500** can be considered the first step up **148**A. Accordingly, the first ramp **146**A can provide a gradual transition from the first step up **148**A to the first land **142**A. Beside significantly reducing the overall length of the expansion toolhead **500**, the risers **148**A, **148**B and **148**C can serve as a natural stop mechanism to inhibit over insertion of the toolhead **500** into the end of PEX tubing during expansion operations, thereby inhibiting inadvertent overexpansion of the and of the PEX tubing. A similar two-step embodiment is depicted in FIGS. **7**C-E.

[0059] In other embodiments, the gradual expansion of toolheads **300**, **400** can be utilized to gradually expand a smaller size of PEX tubing to accommodate a larger size fitting. For example, the end of a ⅜ inch standard sized PEX tubing could be gradually expanded by first inserting the first land **142**A into the PEX tubing to allow for normal expansion, followed by the gradual insertion of the ramp **146**B leading up to the second land **142**B, which in one embodiment could be sized for the normal expansion of ½ inch PEX tubing. Accordingly, through this method a section of ⅜ inch PEX tubing could be directly connected to a ½ inch fitting. Other combinations of sizing expansions are also contemplated.

[0060] Embodiments of the stepped expansion toolhead **100**, **200**, **300**, **400** and **500** can be operably coupled to expansion tools. For example, as depicted in FIGS. **8**A-D, the stepped expansion toolhead **100** can be operably coupled to a manual expansion tool **602** (e.g., manufactured by Iwiss®, Ridgid® or Uponor®), or a powered expansion tool **604**A, **604**B and **604**C (e.g., manufactured by Milwaukee Tool®, DeWalt®, or Uponor®).

[0061] Alternatively, stepped expansion toolheads disclosed herein can be fixedly coupled to an expansion tool as an integral component, thereby potentially eliminating the need to create a threaded coupling between the expansion toolhead and the expansion tool, as well as knurling of the cap **102**, which is common among such devices. In particular, where a threaded coupling is utilized, generally the cap must have an overall length sufficient to include at least three consecutive threads to ensure that the toolhead remains coupled to the expansion tool during operation. Additionally, typically power tools of this nature include a course threading, which is wider than fine threading, but is less susceptible to cross-threading or stripping in the often dirty environment of a construction site.

[0062] For example, as depicted in FIGS. **9**A and **9**B, the expansion toolhead **100** can be fixedly coupled to an expansion tool **704**A, **704**B via a C-clip **706** or other coupling mechanism. Accordingly, including the expansion toolhead **100** as an integral component of the expansion tool **704** can serve to reduce the overall length of the toolhead **100**, thereby contributing to a weight and bulk savings which is advantageous in both the reduction of manufacturing costs and promoting a device which is less bulky and easier to use.

[0063] Various embodiments of systems, devices, and methods have been described herein. These embodiments are given only by way of example and are not intended to limit the scope of the claimed inventions. It should be appreciated, moreover, that the various features of the embodiments that have been described may be combined in various ways to produce numerous additional embodiments. Moreover, while various materials, dimensions, shapes, configurations and locations, etc. have been described for use with disclosed embodiments, others besides those disclosed may be utilized without exceeding the scope of the claimed inventions.

[0064] Persons of ordinary skill in the relevant arts will recognize that the subject matter hereof may comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features of the subject matter hereof may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, the various embodiments can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art. Moreover, elements described with respect to one embodiment can be implemented in other embodiments even when not described in such embodiments unless otherwise noted.

[0065] Although a dependent claim may refer in the claims to a specific combination with one or more other claims, other embodiments can also include a combination of the dependent claim with the subject matter of each other dependent claim or a combination of one or more features with other dependent or independent claims. Such combinations are proposed herein unless it is stated that a specific combination is not intended.

[0066] Any incorporation by reference of documents above is limited such that no subject matter is incorporated that is contrary to the explicit disclosure herein. Any incorporation by reference of documents above is further limited such that no claims included in the documents are incorporated by reference herein. Any incorporation by reference of documents above is yet further limited such that any definitions provided in the documents are not incorporated by reference herein unless expressly included herein.

[0067] For purposes of interpreting the claims, it is expressly intended that the provisions of 35 U.S.C. § 112(f) are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1. An expansion toolhead for an expansion tool, comprising: a cap operably coupleable to the expansion tool; a plurality of jaws forming a 360 degree jaw section operably coupled to the cap, each of the plurality of jaws including an expansion tool interface portion forming a conical recess enabling a cone-shaped wedge of the expansion tool to move the jaw section between an unexpanded position and an expanded position, and a cross-linked polyethylene (PEX) tubing interface portion having a distal end, a proximal end and a plurality of lands positioned therebetween configured to form a corresponding plurality of substantially uniform diameter portions of the jaw section sequentially increasing in diameter between the distal end and the proximal end for the expansion of multiple different standard sized diameters of PEX tubing.